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Urban Analysis of Addis Ababa Using Remote Sensing and Machine Learning

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Introduction

- Urbanization is one of the fastest growing global challenges, especially in developing cities such as Addis Ababa.
- Rapid urban expansion affects vegetation cover, infrastructure, environmental quality, and land use patterns.
- Traditional field-based urban monitoring methods are expensive and time consuming.
- Remote sensing and geospatial technologies provide efficient tools for analyzing urban growth using satellite imagery.
- In this project, Landsat satellite imagery and machine learning techniques were used to analyze urban development patterns in Addis Ababa.

Background of the Problem

Addis Ababa has experienced rapid population growth and urban expansion over the last decades.

Major challenges include:

- Reduction of vegetation and green spaces
- Expansion of built-up areas
- Increased pressure on transportation networks
- Environmental degradation and land cover change

Understanding these spatial changes is important for sustainable urban planning and environmental management.

Objectives of the Study

The main objectives of this project are:

- To analyze urban expansion in Addis Ababa using Landsat imagery
- To compute NDVI and NDBI indices
- To integrate raster and vector geospatial datasets
- To perform spatial analysis techniques such as buffering and overlay analysis
- To apply machine learning methods for clustering and classification
- To identify spatial patterns of urban growth

Study Area Description

Addis Ababa is the capital city of Ethiopia and the largest urban center in the country.

- Latitude: 9.03° N
- Longitude: 38.74° E
- Elevation: Approximately 2355 meters above sea level

The city experiences rapid urban growth due to population increase, infrastructure development, and economic activities.

The project uses both raster and vector geospatial datasets.

Raster Data

- Landsat 8/9 Surface Reflectance imagery
- Band 4 (Red)
- Band 5 (Near Infrared)
- Band 6 (Short Wave Infrared)

Vector Data

- Administrative boundary shapefile of Addis Ababa
- Road network data of Ethiopia

The datasets used in this project were collected from the following sources:

- USGS EarthExplorer for Landsat imagery
- GADM database for administrative boundaries
- OpenStreetMap or Ethiopian GIS sources for road network data

These datasets provide accurate and up-to-date spatial information for urban analysis.

Tools and Software Used

The project was implemented using Python and geospatial analysis libraries.

- Google Colab
- GeoPandas
- Rasterio
- NumPy and Pandas
- Matplotlib
- Scikit-learn
- Rasterstats
- PySAL

These tools support raster processing, vector analysis, machine learning, and spatial visualization.

Data Preprocessing

Several preprocessing steps were applied before analysis.

- ① Loading raster and vector datasets
- ② Coordinate system transformation
- ③ Clipping raster to Addis Ababa boundary
- ④ Removing NoData values
- ⑤ Handling missing values
- ⑥ Data integration and normalization

Data preprocessing improves data quality and ensures compatibility between datasets.

Coordinate System Transformation

All datasets were transformed into a common coordinate reference system:

WGS 84 / UTM Zone 37N (EPSG:32637)

Using a common CRS is important because spatial analysis requires all layers to align geographically.

Without CRS transformation:

- Raster and vector data may not overlap
- Spatial analysis results become incorrect
- Clipping and overlay operations may fail

NDVI Calculation

The Normalized Difference Vegetation Index (NDVI) was used to analyze vegetation health and density.

$$NDVI = \frac{B5 - B4}{B5 + B4} \quad (1)$$

Where:

- B5 = Near Infrared Band
- B4 = Red Band

Interpretation:

- High NDVI values indicate healthy vegetation
- Low NDVI values indicate urban or barren areas

NDBI Calculation

The Normalized Difference Built-up Index (NDBI) was used to identify urban and built-up areas.

$$NDBI = \frac{B6 - B5}{B6 + B5} \quad (2)$$

Where:

- B6 = Short Wave Infrared Band
- B5 = Near Infrared Band

Interpretation:

- Positive values indicate built-up areas
- Negative values indicate vegetation or water bodies

Spatial Analysis Techniques

Several spatial analysis methods were applied.

1. Spatial Querying

- Selected roads located within Addis Ababa

2. Buffer Analysis

- Created road buffers to study urban growth near transportation corridors

3. Overlay Analysis

- Combined NDVI, NDBI, and road layers

These methods help understand spatial relationships between urban growth and infrastructure.

Machine learning techniques were applied to classify land cover patterns.

K-Means Clustering

- Grouped pixels with similar spectral characteristics
- Identified urban and non-urban clusters
- Detected spatial hotspots and patterns

Machine learning improves automated urban classification and spatial pattern recognition.

Visualization Techniques

Different visualization methods were used to represent results.

- NDVI vegetation maps
- NDBI built-up maps
- Choropleth maps
- Heatmaps
- Urban cluster maps

Visualization helps communicate spatial patterns and urban growth trends effectively.

Results and Findings

The analysis produced several important findings.

- Urban expansion is concentrated around major roads
- Vegetation cover is decreasing in central urban areas
- High NDBI values correspond to dense built-up regions
- NDVI highlights remaining green spaces and vegetation patches

The results demonstrate the usefulness of remote sensing for urban monitoring.

The integration of geospatial data and machine learning provided detailed insights into urban growth patterns.

Key observations include:

- Transportation infrastructure strongly influences urban expansion
- Urban growth reduces vegetation coverage
- Machine learning clustering effectively separates land cover classes

These findings can support sustainable urban planning and environmental management.

Challenges Encountered

Several technical challenges were encountered during the project.

- Raster and shapefile overlap problems
- Coordinate system mismatches
- Handling missing and NoData values
- Large raster processing limitations

These issues were solved through CRS transformation, raster validation, and data cleaning.

This project successfully demonstrated the use of:

- Remote sensing
- GIS techniques
- Spatial analysis
- Machine learning

for analyzing urban growth in Addis Ababa.

NDVI and NDBI effectively identified vegetation and built-up areas, while clustering techniques helped classify spatial patterns.

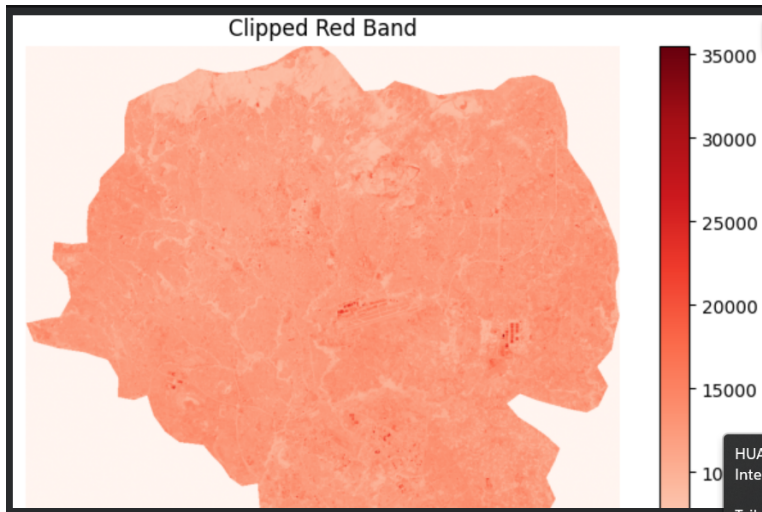
Recommendations

Future improvements may include:

- Using multi-year Landsat images for temporal analysis
- Incorporating population density datasets
- Applying deep learning methods for classification
- Developing web-based GIS dashboards
- Using higher-resolution satellite imagery

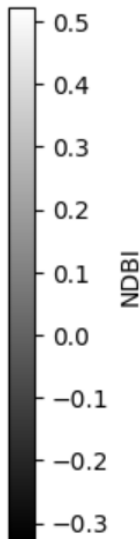
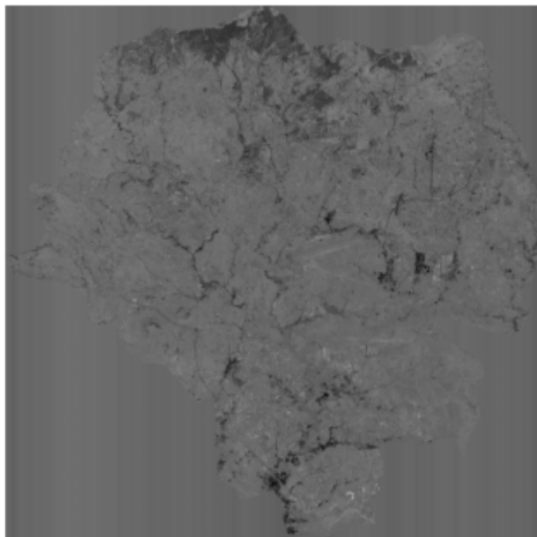
These enhancements can improve urban monitoring accuracy and decision making.

Sample images from the code

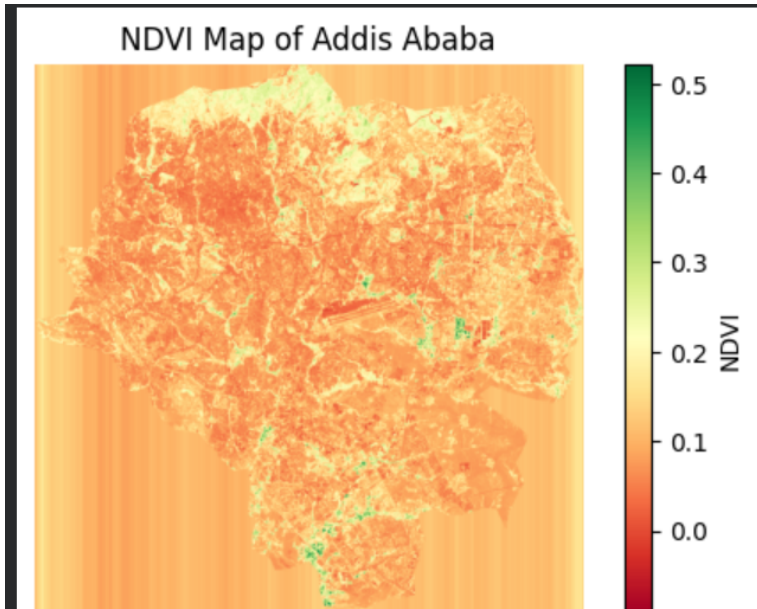


Sample images from the code

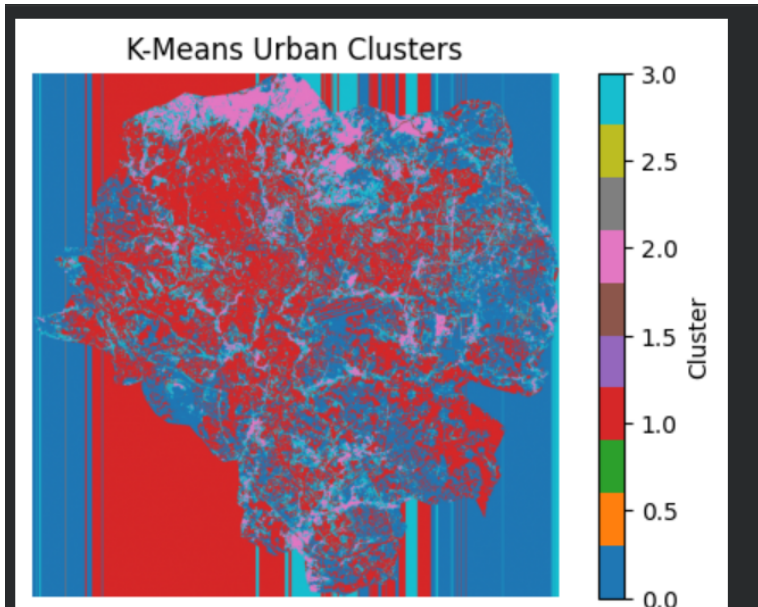
Built-up Area Map



Sample images from the code



Sample images from the code



References

- USGS EarthExplorer: <https://earthexplorer.usgs.gov>
- GeoPandas Documentation: <https://geopandas.org>
- Rasterio Documentation: <https://rasterio.readthedocs.io>
- Scikit-learn Documentation: <https://scikit-learn.org>
- PySAL Documentation: <https://pysal.org>
- Landsat Science: <https://landsat.gsfc.nasa.gov>

Thank You